

HO�F–RIESZ: Round 003

The initial iid pair and collision interface

Campaign research memorandum

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Published input checkpoint: 2f532ccd2cb5f3d84db456be96e438f03f7e2ad2. The initial pair constructions used isolated worktrees based on the published Round 001 input; later proofs and reviews used the Round 002 checkpoint above. This memorandum records initial iid statements. The round report gives the actual per-claim audit status. Neither the bare potential nor its truncated version is identified with the backward corrector, and no positive-time interacting-law estimate is inferred.

1 An exact finite-particle projection formula

Let $N \geq 2$, let $X_1, \dots, X_N$ be iid with probability law μ , and let Φ be real, symmetric and in $L^2(\mu \otimes \mu)$. Set

$$\theta = \mu^2(\Phi), \quad h(x) = \int \Phi(x, y) d\mu(y) - \theta, \quad H(x, y) = \Phi(x, y) - \theta - h(x) - h(y).$$

The canonical kernel H has conditional mean zero in either variable. Orthogonality gives

$$\|\Phi\|_2^2 = \theta^2 + 2\|h\|_2^2 + \|H\|_2^2.$$

With the campaign’s ordered distinct-label denominator N^2 , define

$$P_N[\Phi] = \frac{1}{2N^2} \sum_{i \neq j} \Phi(X_i, X_j) - \frac{1}{N} \sum_i \int \Phi(X_i, y) d\mu(y) + \frac{\theta}{2}.$$

Collecting constants and single-label terms gives the exact identity

$$P_N[\Phi] = -\frac{\theta}{2N} - \frac{1}{N^2} \sum_i h(X_i) + \frac{1}{N^2} \sum_{i < j} H(X_i, X_j). \tag{1}$$

The first projection remains because each label occurs only $N - 1$ times in the unordered pair sum. Conditional centering makes distinct canonical pair terms orthogonal, including pairs sharing one label; it also makes them orthogonal to the linear sum. Therefore

$$\mathbb{E}P_N[\Phi] = -\frac{\theta}{2N}, \tag{2}$$

$$\mathbb{E}|P_N[\Phi]|^2 = \frac{\theta^2}{4N^2} + \frac{\|h\|_2^2}{N^3} + \frac{N-1}{2N^3} \|H\|_2^2 \tag{3}$$

$$\leq \frac{N-1}{2N^3} \|\Phi\|_2^2. \tag{4}$$

The last constant is attained by a nonzero canonical kernel whenever the probability space admits one; at $N = 2$ equality holds for every kernel. A constant kernel yields a deterministic $-\theta/(2N)$, not zero. A kernel $h(x) + h(y)$ with centered h yields precisely the linear term in (1). These test the centering and the denominator.

There is no self-label evaluation $\Phi(X_i, X_i)$. For atomless μ , spatial-diagonal values may be changed freely. With atoms, distinct labels can coincide spatially, and diagonal values at those atoms belong to the $L^2(\mu^2)$ equivalence class. Deleting labels does not delete these values.

2 Riesz normalization and the square-integrability boundary

Now use Haar probability measure on the unit torus and $0 < s < d$. Write

$$c_{d,s} = \pi^{s-d/2} \frac{\Gamma((d-s)/2)}{\Gamma(s/2)}, \quad \widehat{g}(k) = c_{d,s} |k|^{s-d} \quad (k \neq 0), \quad \widehat{g}(0) = 0.$$

Let p_t be the torus heat kernel with multiplier $e^{-4\pi^2 t |k|^2}$. Its Gaussian periodization gives the L^1 representation

$$g(x) = \frac{4^{(d-s)/2} \pi^{d/2}}{\Gamma(s/2)} \int_0^\infty t^{(d-s)/2-1} (p_t(x) - 1) dt. \quad (5)$$

At small t the L^1 norm of $p_t - 1$ is at most two, and at large t it decays exponentially. Fubini computes exactly the prescribed Fourier coefficient. The whole-space Gaussian term in (5) integrates to $|x|^{-s}$ with coefficient one. Nonzero lattice translates give integrable $e^{-c/t}$ bounds near $t = 0$, with all derivatives; the large-time difference is also smooth. Hence locally

$$g(x) = |x|^{-s} + H_0(x), \quad H_0 \text{ smooth near } 0.$$

In particular g is integrable and bounded below, and belongs to L^2 exactly when $2s < d$.

For Haar iid samples the pair kernel $g(x - y)$ is canonical in L^1 , so

$$P_N[g] = N^{-2} \sum_{i < j} g(X_i - X_j), \quad \mathbb{E} P_N[g] = 0.$$

It is finite almost surely for every finite N . If $2s \geq d$, its second moment is infinite. Indeed, a lower bound $g \geq -C$ implies, with $M = N(N-1)/2$,

$$(P_N[g])_+ \geq N^{-2} (g(X_1 - X_2) - C(M-1))_+,$$

whose square has infinite expectation. This argument does not expand undefined cross moments and does not assert nonconvergence in probability.

3 Heat cutoffs and two different spatial scales

Let $g_\varepsilon = p_\varepsilon * g$ and $b_N = \min(\beta_N, 1)$, $\sigma_N = \sqrt{N b_N}$. Parseval and (3) give

$$\|g_\varepsilon\|_2^2 = c_{d,s}^2 \sum_{k \neq 0} |k|^{2s-2d} e^{-8\pi^2 \varepsilon |k|^2}, \quad (6)$$

$$\mathbb{E} |\sigma_N P_N[g_\varepsilon]|^2 = \frac{b_N(N-1)}{2N^2} \|g_\varepsilon\|_2^2. \quad (7)$$

Counting lattice points in max-norm shells in (6) gives, with positive constants depending on d, s and a fixed small cutoff range,

$$\|g_\varepsilon\|_2^2 \asymp \begin{cases} 1, & 2s < d, \\ \log(1/\varepsilon), & 2s = d, \\ \varepsilon^{-(2s-d)/2}, & 2s > d. \end{cases}$$

The shell $|k|_\infty = n$ has $(2n+1)^d - (2n-1)^d$ points, between $2dn^{d-1}$ and $2d3^{d-1}n^{d-1}$. On this shell $n \leq |k| \leq \sqrt{d}n$, so the sum is bounded above and below by explicit constants times $\sum_{n \geq 1} n^{2s-d-1} e^{-a\varepsilon n^2}$, with $a = 8\pi^2$ and $8\pi^2 d$, respectively. Integral comparison gives the displayed three orders. The full proof in the sealed pair memorandum gives explicit two-sided constants and their cutoff interval.

At $\varepsilon_N = N^{-2/d}$, the high-singularity scaled second moment is of order $b_N N^{2s/d-2}$ and tends to zero for every $s < d$. At $\varepsilon_N = N^{-4/d}$ it is of order $b_N N^{4s/d-3}$ when $2s > d$. These cutoffs correspond respectively to lengths $N^{-1/d}$ and $N^{-2/d}$. A regularized variance tending to zero does not control the unregularized statistic: for $2s \geq d$ the latter and its difference from every fixed smooth cutoff have infinite second moment. A probability comparison needs a separate estimate.

4 A probability estimate that retains close pairs and the mean

Fix $0 < r < 1/4$. Let

$$h_r = g \mathbf{1}_{\{\text{dist}(x,0) \geq r\}}, \quad \tau_r = \int_{\text{dist}(x,0) < r} g(x) dx, \quad k_r = h_r + \tau_r.$$

Then k_r is canonical. Let E_r be the event that no pair of distinct labels is within distance r , and let v_d be the unit-ball volume. A union bound and the exact normalization give

$$\mathbb{P}(E_r^c) \leq \frac{N(N-1)}{2} v_d r^d, \quad P_N[g] = P_N[k_r] - \frac{N-1}{2N} \tau_r \quad \text{on } E_r.$$

The mean correction has coefficient $(N-1)/(2N)$, not zero. Put

$$V_r = \|k_r\|_2^2, \quad B_N(r) = \sigma_N \frac{N-1}{2N} |\tau_r|.$$

For $z > B_N(r)$, Chebyshev and (7) for this canonical kernel yield

$$\mathbb{P}\{|\sigma_N P_N[g]| > z\} \leq \frac{N(N-1)}{2} v_d r^d + \frac{b_N(N-1)V_r}{2N^2(z - B_N(r))^2}. \quad (8)$$

The local singularity supplies

$$\tau_r = \frac{dv_d}{d-s} r^{d-s} + O(r^d), \quad V_r \leq C_{d,s} \begin{cases} 1, & 2s < d, \\ 1 + \log(1/r), & 2s = d, \\ r^{d-2s}, & 2s > d. \end{cases}$$

The constants may equivalently be expressed by the displayed smooth remainder and the integral outside the fixed ball; they are independent of N, β_N, r .

At $2s = d$, choose $r_N = N^{-3/d}$. For $b_N = 1$, close-pair probability, scaled discarded mean, and truncated variance are respectively $O(N^{-1})$, $O(N^{-1})$, and $O(\log N/N)$. For $d/2 < s < 3d/4$, choose

$$r_N = N^{-a}, \quad \frac{2}{d} < a < \frac{1}{2s-d}.$$

All three errors in (8) then vanish. Together with the direct L^2 result below $d/2$, this proves

$$0 < s < \frac{3d}{4} \implies \sqrt{N}P_N[g] \longrightarrow 0 \quad \text{in probability.}$$

It also holds at $\sigma_N \leq \sqrt{N}$ for every positive temperature sequence in this range.

For $d/2 < s < d$ a further sufficient condition is

$$b_N N^{4s/d-3} \longrightarrow 0.$$

Indeed put $A_N = \sqrt{b_N} N^{2s/d-3/2}$ and choose $r_N = N^{-2/d} A_N^{1/(2s-d)}$. The three controlling factors are

$$N^2 r_N^d = A_N^{d/(2s-d)}, \quad \sigma_N r_N^{d-s} = A_N^{s/(2s-d)}, \quad \frac{b_N}{N} r_N^{d-2s} = A_N.$$

This proves the asserted probability convergence. It supplies no converse. At $s \geq 3d/4$ and $b_N = 1$, requiring no close pairs forces $r_N = N^{-2/d} \ell_N$ with $\ell_N \rightarrow 0$, whereas the scaled truncated variance grows as $N^{4s/d-3} \ell_N^{-(2s-d)}$. This disproves closure of that particular truncation strategy, not convergence of the original statistic. An endpoint or stable-law claim would require a new proof.

5 An absolute-moment estimate and a sharp converse

Center both parts of the spatial split: $k_r = g \mathbf{1}_{\{\rho \geq r\}} + \tau_r$ and $j_r = g \mathbf{1}_{\{\rho < r\}} - \tau_r$, where ρ denotes distance to zero. Then $g = k_r + j_r$. Put $D(r) = \int_{\rho < r} |g| \leq C_1 r^{d-s}$. The exact outer variance and the triangle inequality over the unordered inner sum give

$$\mathbb{E}|\sigma_N P_N[g]| \leq \sqrt{\frac{b_N}{2N}} \|g \mathbf{1}_{\{\rho \geq r\}}\|_2 + \sqrt{N b_N} D(r).$$

Indeed $\|j_r\|_1 \leq 2D(r)$ and $\mathbb{E}|P_N[j_r]| \leq (N-1)\|j_r\|_1/(2N)$. Choose $r = N^{-2/d}$. For fixed d, s this proves, uniformly in every positive temperature,

$$\mathbb{E}|\sigma_N P_N[g]| \leq C_{d,s} \begin{cases} \sqrt{b_N/N}, & 2s < d, \\ \sqrt{b_N(1 + \log N)/N}, & 2s = d, \\ \sqrt{b_N} N^{2s/d-3/2}, & 2s > d. \end{cases} \quad (9)$$

The first row follows directly from the untruncated second moment. No exclusion of close pairs is used in this argument. It was constructed separately in the statement-only probability reconstruction.

Here is the new converse for $d/2 < s < d$. Let $u_N = N^{2s/d-2}$ and $R_N = N^{-2/d}$. Fix a sufficiently small radius R such that $g(x) \geq \frac{1}{2}|x|^{-s}$ for $0 < |x| < R$, $\tau_r \leq C_\tau r^{d-s}$ and $V_r \leq C_V r^{d-2s}$ for $r < R$. For large N take $R_N < R$ and count pairs at distance less than δR_N , with a fixed $0 < \delta < 1$:

$$Z_N = \sum_{i < j} \mathbf{1}_{\{|X_i - X_j| < \delta R_N\}}, \quad m_N = \mathbb{E}Z_N = \frac{N(N-1)}{2} v_d \delta^d N^{-2}.$$

Distinct pair indicators are pairwise independent: for shared labels, condition on that label and use Haar translation invariance. Joint independence of all the indicators is neither true in general nor used. Thus $\mathbb{E}Z_N^2 \leq m_N + m_N^2$, and Cauchy-Schwarz yields

$$\mathbb{P}(Z_N > 0) \geq \frac{m_N}{1 + m_N} \geq c_0 \delta^d, \quad c_0 = \frac{v_d}{4(1 + v_d/2)}.$$

The exact outer-inner decomposition is

$$P_N[g] = P_N[k_{R_N}] + N^{-2} \sum_{i < j} g(X_i - X_j) \mathbf{1}_{\{|X_i - X_j| < R_N\}} - \frac{N-1}{2N} \tau_{R_N}.$$

All inner summands are positive. On $Z_N > 0$ at least one contributes $\frac{1}{2} \delta^{-s} u_N$. If $\delta^{-s} \geq 2C_\tau$, the last two terms are at least $\frac{1}{4} \delta^{-s} u_N$. Meanwhile

$$\mathbb{E}P_N[k_{R_N}]^2 \leq \frac{C_V}{2} u_N^2, \quad \mathbb{P}(P_N[k_{R_N}] < -\frac{1}{8} \delta^{-s} u_N) \leq 32C_V \delta^{2s}.$$

Since $2s > d$, choose the same fixed δ so that $\delta^{2s-d} \leq c_0/(64C_V)$. Subtracting the last failure probability from the close-pair probability, without assuming event independence, proves

$$\mathbb{P}(P_N[g] \geq a u_N) \geq p_0, \quad a = \delta^{-s}/8, \quad p_0 = c_0 \delta^d/2. \quad (10)$$

Consequently $a p_0 u_N \leq \mathbb{E}|P_N[g]| \leq C_{d,s} u_N$. Multiplying by σ_N shows that, in this range, vanishing in probability is equivalent to vanishing in L^1 , and equivalent to

$$B_N := b_N N^{4s/d-3} \longrightarrow 0.$$

For necessity, a subsequence on which $B_N \geq \eta > 0$ retains the fixed positive tail probability in (10). Bounded B_N gives tightness by the absolute-moment upper bound. From an unbounded scalar subsequence extract a further one tending to infinity; the same lower probability then proves non-tightness. At $\sqrt{N}$ scale the bare statistic is therefore negligible below $3d/4$, tight and nonvanishing at $3d/4$, and not tight above it. No endpoint distribution or stable law is identified. This converse is a separate assertion from the earlier sufficient-bound argument; the round report records its actual independent review status.

6 An exact model retaining internal pair transport

The raw-potential threshold cannot automatically be applied to the backward corrector. A solvable model makes the distinction concrete. Let A be a fixed symmetric matrix, $z = r\theta \neq 0$, $p = s + 2$ and $\tau = T - t$. Retain only the Euclidean internal pair drift:

$$\left(\partial_t + \frac{2s}{N} r^{-s-2} z \cdot \nabla_z \right) \Phi_N = -s r^{-s} \theta^T A \theta, \quad \Phi_N(T, z) = 0.$$

The factor two comes from $\nabla_x - \nabla_y = 2\nabla_z$. Directions are constant along characteristics and $r(h)^p = r^p + 2sph/N$. Integrating the source along the forward characteristic gives the unique solution in the absolutely continuous characteristic class, with no equation or boundary condition imposed at zero:

$$\Phi_N(t, z) = \frac{N}{4} (\theta^T A \theta) \left[(r^p + 2sp\tau/N)^{2/p} - r^2 \right]. \quad (11)$$

Differentiation gives $-\partial_\tau F + (2s/N)r^{-s-1}\partial_r F = -sr^{-s}$ for the radial amplitude F , checking both sign and coefficient.

Set $\ell = (2sp\tau/N)^{1/p}$ for $\tau > 0$. Then $F = (N/4)\ell^2 H_s(r/\ell)$, where

$$H_s(u) = (1 + u^p)^{2/p} - u^2 = \frac{2}{p} \int_0^1 (u^p + v)^{-s/p} dv.$$

This decreases from one. It is bounded above and below by positive constants on $[0, 1]$, and by constants times u^{-s} on $[1, \infty)$. The exact angular factor is

$$Q(A) = \int_{S^{d-1}} (\theta^T A \theta)^2 d\theta = \frac{|S^{d-1}|}{d(d+2)} [(\text{tr } A)^2 + 2 \text{tr}(A^2)].$$

Thus every nonzero traceless symmetric A also has a positive square norm. For any fixed ball,

$$\|\Phi_N(t)\|_{L^2(B_R)}^2 = \frac{Q(A)N^2\ell^{d+4}}{16} \int_0^{R/\ell} H_s(u)^2 u^{d-1} du.$$

When $\ell \leq R$, the three orders are respectively

$$Q(A) \begin{cases} \tau^2 R^{d-2s}, & 2s < d, \\ \tau^2 [1 + \log(R/\ell)], & 2s = d, \\ N^{(2s-d)/(s+2)} \tau^{(d+4)/(s+2)}, & 2s > d. \end{cases}$$

The full sealed proof supplies two-sided constants, the $\ell > R$ case, and the zero-time endpoint. Despite finite radial amplitude, a nonscalar A has direction-dependent limits at collision. No smooth extension is available in that case.

Choose a fixed smooth radial cutoff supported inside a torus ball of radius less than $1/2$ and define the periodic pair diagnostic $\mathcal{H}_{N,\tau}(x, y) = \chi(x - y)\Phi_N(T - \tau, x - y)$. It is an L^2 kernel; its diagonal value is immaterial for a density. For iid preparation with density bounded by M ,

$$\|\mathcal{H}_{N,\tau}\|_{L^2(\mu^2)}^2 \leq M \|\chi\Phi_N\|_2^2.$$

Equation (4) now retains all centering contributions, and yields

$$\sup_{0 \leq \tau \leq T} \mathbb{E} |\sigma_N P_N[\mathcal{H}_{N,\tau}]|^2 \leq C_T b_N \begin{cases} N^{-1}, & 2s < d, \\ (1 + \log N)/N, & 2s = d, \\ N^{-(d+2-s)/(s+2)}, & 2s > d. \end{cases}$$

Every row vanishes for $s < d$. The supremum is outside the expectation. This is an initial iid diagnostic for a zero-diffusion equation; positive temperature appears only in the statistic's scale.

7 What remains load-bearing

These estimates hold at initial iid Haar preparation for the raw potential. The probability result beyond the L^2 range illustrates why variance divergence is not a probability obstruction. It does not identify the backward pair kernel, compare singular and regularized particle dynamics, or control any positive-time martingale. The sharp converse concerns only the raw potential. The transport diagnostic, also available for bounded densities, demonstrates a different initial endpoint power and does not solve the full pair equation.

For the genuine pair corrector Ψ_N , a sufficient remaining initial comparison is

$$\sqrt{b_N/N} \|\Psi_N(0) - \mathcal{H}_{N,T}\|_{L^2(\mu_0^2)} \longrightarrow 0.$$

This is open. The error equation must retain $2\nu\Delta_z$, ordinary transport, both nonlocal response terms, the actual test/source and periodic force remainder, and the cutoff term

$$(2s/N)r^{-s-1}\chi'(r)\Phi_N.$$

For a nonscalar A , the leading angular Laplacian has a term proportional to r^{-2} , so diffusion cannot be added by an unproved small-coefficient argument. Well-posedness, cutoff passage, evolved-law residuals and martingale estimates also remain load-bearing. A weaker topology could replace the displayed comparison only with its own proved fluctuation-scale estimate.

The separate transport review makes one limitation precise: if the traceless part of A is nonzero and $\tau > 0$, the squared punctured Laplacian contains a positive multiple of $\int_0^\epsilon r^{d-5} dr$. Thus it is not locally L^2 in dimensions 2, 3, 4; in dimension two its absolute L^1 norm also diverges. This disproves a proposed separate-norm perturbation estimate for this profile, not existence of a different diffusion-retaining solution.

There is also an exact punctured rescaling of the local equation with diffusion retained. Set $y = N^{1/(s+2)}z$ and $\Phi_N = N^{s/(s+2)}F_N(\tau, y)$. The chain rule gives

$$\partial_\tau F_N = 2\chi_N \Delta_y F_N + 2s|y|^{-s-2}y \cdot \nabla_y F_N + s|y|^{-s}a(y/|y|), \quad \chi_N = \frac{N^{2/(s+2)}}{\beta_N}.$$

At critical microscopic coupling the exact identity is

$$\chi_N = \lambda_N^{-1} N^{s(s+2-d)/(d(s+2))}.$$

Consequently this coefficient tends to zero, a positive constant, or infinity according as $d > s + 2$, $d = s + 2$, or $d < s + 2$. This is only an operator rescaling on the punctured domain. No collision domain, solution convergence, or limit exchange follows from the coefficient limit. A future local comparison must handle its degree-zero and degree-two angular modes in the appropriate parameter range.

The three regimes remain distinct:

$$\beta_N N^{2s/d-1} \rightarrow 0, \quad \lambda_N = \beta_N N^{s/d-1} \rightarrow 0, \quad \lambda_N \rightarrow \lambda \in (0, \infty).$$

They are respectively the old energy-floor condition, full microscopic subcriticality, and microscopic criticality. None is replaced by the sufficient initial-pair condition above. The logarithmic interaction remains outside this positive-power calculation.